

CHAPTER 4 : ENERGY ANALYSIS OF CLOSED SYSTEM

Section	Core Revision Focus
4.1	Examines how energy is transferred through the expansion and compression of system boundaries.
4.2	Establishes how conservation of energy governs processes in closed thermodynamic systems.
4.3	Introduces specific heats as measures of how thermodynamic energy properties vary with temperature.
4.4	Develops the temperature-based energy-property relations used to analyze ideal gases.
4.5	Applies energy-property relations to substances whose volume changes are negligible.

4.1 MOVING BOUNDARY WORK

CORE RELATIONS

Quasi-equilibrium process: system remains nearly in equilibrium at every instant; a continuous state path and well-defined P - V relation exist.

For a piston-cylinder:

$$\delta W_b = F ds = PA ds = P dV$$

$$W_b = \int_1^2 P dV$$

- P = absolute pressure.
- Expansion: $dV > 0 \Rightarrow W_b > 0$ (work by system).
- Compression: $dV < 0 \Rightarrow W_b < 0$ (work on system).
- Boundary work is a **path function**.

For nonquasi-equilibrium motion:

$$W_b = \int_1^2 P_i dV$$

where P_i is the pressure acting on the inner face of the piston.

GRAPHICAL INTERPRETATION

For a quasi-equilibrium process:

$$dA = P dV$$

$$A = \int_1^2 P dV = W_b$$

- Area under a P - V process curve $\rightarrow$ magnitude of boundary work.
- Area on a P - v diagram $\rightarrow$ boundary work per unit mass.
- For a cycle, enclosed P - V area $\rightarrow$ magnitude of net work.

$$W_{\text{net}} = W_{\text{expansion}} - W_{\text{compression}}$$

EXPERIMENTAL AND CONSTANT-VOLUME CASES

If $P = f(V)$ is unavailable analytically but experimental P - V data are known:

1. plot the data,
2. construct the process curve,

3. determine the area graphically.

Constant volume:

$$dV = 0$$

$$\omega_b = 0$$

CONSTANT-PRESSURE PROCESS

$$\omega_b = P_0(V_2 - V_1)$$

Using $V = mv$:

$$\omega_b = mP_0(v_2 - v_1)$$

ISOTHERMAL IDEAL-GAS PROCESS

For constant T:

$$PV = mRT_0 = C$$

$$P = \frac{C}{V}$$

Hence,

$$\omega_b = C \ln \left(\frac{V_2}{V_1} \right)$$

$$\omega_b = P_1 V_1 \ln \left(\frac{V_2}{V_1} \right)$$

For an ideal gas:

$$P_1 V_1 = P_2 V_2 = mRT_0$$

$$\frac{V_2}{V_1} = \frac{P_1}{P_2}$$

Compression $\Rightarrow V_2 < V_1 \Rightarrow \omega_b < 0$.

POLYTROPIC PROCESS

$$PV^n = C$$

For $n \neq 1$:

$$\omega_b = \frac{P_2 V_2 - P_1 V_1}{1 - n}$$

For an ideal gas:

$$\omega_b = \frac{mR(T_2 - T_1)}{1 - n}$$

EXPANSION AGAINST A LINEAR SPRING

Spring force:

$$F = kx$$

Spring pressure:

$$P_{\text{spring}} = \frac{F}{A}$$

Since spring force varies linearly with displacement, the P - V path is a straight line.

Spring work:

$$W_{\text{spring}} = \frac{1}{2}k(x_2^2 - x_1^2)$$

Total work:

$$W_{\text{total}} = P_1(V_2 - V_1) + \frac{1}{2}k(x_2^2 - x_1^2)$$

ENERGY DESTINATION OF BOUNDARY WORK

Boundary work transfers energy to other mechanisms, e.g.

$$W_b = W_{\text{friction}} + W_{\text{atm}} + W_{\text{crank}}$$

4.2 ENERGY BALANCE FOR CLOSED SYSTEMS

GENERAL ENERGY BALANCE

Conservation of energy:

$$E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{system}}$$

Total system energy may include internal, kinetic, potential, and other forms.

Rate form:

$$\dot{E}_{\text{in}} - \dot{E}_{\text{out}} = \frac{dE_{\text{system}}}{dt}$$

For constant rates:

$$Q = \dot{Q} \Delta t, \quad W = \dot{W} \Delta t, \quad \Delta E = \frac{dE}{dt} \Delta t$$

Per unit mass:

$$e_{\text{in}} - e_{\text{out}} = \Delta e_{\text{system}}$$

CYCLE AND FIRST-LAW FORMS

For a cycle:

$$\Delta E_{\text{system}} = 0$$

$$E_{\text{in}} = E_{\text{out}}$$

For a closed-system cycle:

$$W_{\text{net,out}} = Q_{\text{net,in}}$$

$$\dot{W}_{\text{net,out}} = \dot{Q}_{\text{net,in}}$$

Classical sign convention:

- $Q > 0$: heat transfer into system.
- $W > 0$: work done by system.

Thus:

$$Q - W = \Delta E$$

where

$$Q = Q_{\text{in}} - Q_{\text{out}}$$

and

$$W = W_{\text{out}} - W_{\text{in}}$$

Negative Q or W means the actual direction is opposite to the assumed positive direction.

STATIONARY CLOSED SYSTEM

If kinetic and potential energy changes are negligible:

$$Q - W = \Delta U$$

Per unit mass:

$$q - w = \Delta e$$

Differential form:

$$\delta q - \delta w = de$$

Heat and work are both **energy-transfer mechanisms**, not properties of the system.

UNRESTRAINED EXPANSION

Expansion into a vacuum:

$$P_{ext} = 0$$

Therefore:

$$W_b = \int_1^2 P_{ext} dV = 0$$

If no other work is present:

$$W = 0$$

For a stationary closed system:

$$Q = \Delta U$$

Important: work depends on the chosen system boundary; including the evacuated space can make the boundary fixed.

4.3 SPECIFIC HEATS

DEFINITIONS

Specific heat = energy required to raise the temperature of unit mass by one degree under a specified condition.

$$c_v = \left(\frac{\partial u}{\partial T} \right)_v$$

$$c_p = \left(\frac{\partial h}{\partial T} \right)_p$$

Hence:

- c_v describes the variation of internal energy with T at constant v .
- c_p describes the variation of enthalpy with T at constant P .
- Generally, $c_p > c_v$.

These are **property relations**; they are not restricted to processes actually occurring at constant v or constant P .

STATE DEPENDENCE AND UNITS

In general, specific heats depend on the state of the substance.

Common units:

$$\frac{\text{kJ}}{\text{kg} \cdot ^\circ\text{C}} \quad \text{or} \quad \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$$

Since temperature intervals have equal magnitude:

$$\Delta T(^{\circ}\text{C}) = \Delta T(\text{K})$$

Molar specific heats:

$$\bar{c}_v, \quad \bar{c}_p$$

4.4 INTERNAL ENERGY, ENTHALPY, AND SPECIFIC HEATS OF IDEAL GASES

IDEAL-GAS RELATIONS

$$Pv = RT$$

For an ideal gas:

$$u = u(T)$$

Enthalpy definition:

$$h = u + Pv$$

Since $Pv = RT$:

$$h = u + RT$$

Therefore:

$$h = h(T)$$

Key result: for an ideal gas, u , h , c_v , and c_p depend only on temperature.

IDEAL-GAS SPECIFIC HEATS

$$c_v = c_v(T)$$

$$c_p = c_p(T)$$

Differential relations:

$$du = c_v(T) dT$$

$$dh = c_p(T) dT$$

Changes between states:

$$\Delta u = \int_{T_1}^{T_2} c_v(T) dT$$

$$\Delta h = \int_{T_1}^{T_2} c_p(T) dT$$

Thus, Δu and Δh depend only on T_1 , T_2 , not on the process path.

AVERAGE SPECIFIC HEAT

For moderate/small temperature ranges:

$$\Delta u \approx c_{v,avg}(T_2 - T_1)$$

$$\Delta h \approx c_{p,avg}(T_2 - T_1)$$

Useful temperature estimate:

$$T_{avg} = \frac{T_1 + T_2}{2}$$

or

$$c_{avg} = \frac{c_1 + c_2}{2}$$

Important:

$$\Delta u = c_{v,avg} \Delta T$$

and

$$\Delta h = c_{p,avg} \Delta T$$

are applicable to **any ideal-gas process**; the subscripts do not specify the process path.

MONATOMIC IDEAL GASES

Specific heats are essentially constant over a wide range:

$$\Delta u = c_v(T_2 - T_1)$$

$$\Delta h = c_p(T_2 - T_1)$$

METHODS FOR DETERMINING Δu AND Δh

1. Property tables

$$\Delta u = u_2 - u_1, \quad \Delta h = h_2 - h_1$$

2. Temperature-dependent specific heats

$$\Delta u = \int_{T_1}^{T_2} c_v(T) dT, \quad \Delta h = \int_{T_1}^{T_2} c_p(T) dT$$

3. Average specific heats

$$\Delta u \approx c_{v,avg} \Delta T, \quad \Delta h \approx c_{p,avg} \Delta T$$

REFERENCE STATE

Reference values of u and h are arbitrary; only differences matter:

$$\Delta u = u_2 - u_1$$

$$\Delta h = h_2 - h_1$$

SPECIFIC-HEAT RELATIONS

From

$$h = u + RT$$

$$dh = du + R dT$$

Therefore:

$$c_p = c_v + R$$

$$c_v = c_p - R$$

On a molar basis:

$$\bar{c}_p = \bar{c}_v + R_u$$

Specific heat ratio:

$$k = \frac{c_p}{c_v}$$

k varies mildly with temperature and may often be treated as constant over a limited temperature range.

4.5 INTERNAL ENERGY, ENTHALPY, AND SPECIFIC HEATS OF SOLIDS AND LIQUIDS

INCOMPRESSIBLE SUBSTANCES

For an incompressible substance:

$$v = \text{constant}$$

or

$$\rho = \text{constant}$$

Solids and liquids are commonly approximated as incompressible.

SPECIFIC HEAT AND INTERNAL ENERGY

For incompressible substances:

$$c_p = c_v = c$$

and

$$c = c(T)$$

Internal-energy change:

$$du = c(T) dT$$

$$\Delta u = \int_{T_1}^{T_2} c(T) dT$$

For approximately constant c :

$$\Delta u \approx c_{\text{avg}}(T_2 - T_1)$$

ENTHALPY OF INCOMPRESSIBLE SUBSTANCES

$$h = u + Pv$$

Since

$$dv = 0$$

then

$$dh = du + v dP$$

Using $du = c(T) dT$:

$$dh = c(T) dT + v dP$$

Thus:

$$\Delta h = \Delta u + v \Delta P$$

Average- c form:

$$\Delta h \approx c_{\text{avg}} \Delta T + v \Delta P$$

$$\Delta h \approx c_{\text{avg}}(T_2 - T_1) + v(P_2 - P_1)$$

SPECIAL CASES

Solids:

$$v\Delta P \approx 0$$

Therefore:

$$\Delta h \approx \Delta u$$

$$\Delta h \approx c_{avg}(T_2 - T_1)$$

Liquid at constant pressure:

$$\Delta P = 0$$

$$\Delta h = \Delta u$$

$$\Delta h \approx c_{avg}(T_2 - T_1)$$

Liquid at constant temperature:

$$\Delta T = 0$$

$$\Delta u = 0$$

$$\Delta h = v\Delta P$$

COMPRESSED LIQUID ENTHALPY

Approximation from saturated-liquid properties at the same temperature:

$$h_{P,T} \approx h_f(T) + v_f(T) [P - P_{sat}(T)]$$

Simpler approximation:

$$h_{P,T} \approx h_f(T)$$

The pressure correction may be neglected when sufficiently small; at high T and P , the simpler approximation can sometimes be preferable.

ENERGY BALANCE FOR MULTIPLE INCOMPRESSIBLE SUBSTANCES

For several substances:

$$\Delta U_{sys} = \sum_i \Delta U_i$$

For an isolated system:

$$\Delta U_{sys} = 0$$

For each substance:

$$\Delta U_i = m_i \Delta u_i$$

With constant specific heats:

$$\Delta U_i \approx m_i c_i (T_2 - T_{1,i})$$

Hence:

$$\sum_i m_i c_i (T_2 - T_{1,i}) = 0$$

For thermal equilibrium:

$$T_1 = T_2 = \dots = T_{eq}$$

Energy lost by hotter substances = energy gained by colder substances.

HEAT TRANSFER FOR INCOMPRESSIBLE SUBSTANCES

For constant c :

$$\Delta U = mc(T_2 - T_1)$$

Cooling:

$$Q_{out} = mc(T_1 - T_2) \quad (T_1 > T_2)$$

Heating:

$$Q_{in} = mc(T_2 - T_1) \quad (T_2 > T_1)$$